

Bridging the Generational Lexicon: Gen Z Slang Translation with Behavioral Consistency Constraints

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Abstract

Generation Z's rapidly evolving digital vernacular presents unique challenges for NLP systems trained on standard English. We address two complementary tasks: (1) classifying dialogue acts and emotions at the utterance level using DailyDialog, and (2) bidirectional translation between Gen Z slang and standard English. We propose an integrated pipeline that generates multiple translation candidates and reranks them by downstream behavioral consistency—whether the translation preserves the source's predicted dialogue act and emotion. Our BART-based translation model, fine-tuned on a combined corpus of multiple Gen Z datasets, achieves ROUGE-1 of 0.81 and BERTScore F1 of 0.98 for Gen Z $\rightarrow$ standard English, outperforming a T5 baseline. We find that emotion classification is severely hampered by extreme class imbalance in DailyDialog, with both BERT and BERTweet defaulting to "no emotion" for most inputs. We analyze this limitation in depth and propose concrete directions for improvement. An interactive Gradio demo visualizes the full pipeline.

1. Introduction

Language is not static. Each generation reshapes vocabulary, grammar, and communicative norms to reflect its cultural and technological context. Generation Z—those born roughly between 1997 and 2012—has developed a particularly distinctive digital vernacular characterized by lexical compression (“rizz” for charisma, “sus” for suspicious), semantic shifts (“fire” meaning excellent, “mid” meaning mediocre), and flexible grammatical usage (“ghosting,” “adulthood”) (Bansal & Mamidi, 2025). These patterns are shaped by communication on platforms such as TikTok, Twitter/X, and Discord, where brevity, irony, and in-group signaling are emphasized.

This evolving form of language presents challenges for natural language processing (NLP) systems. Most models are trained on large corpora of relatively formal or standardized text, which limits their ability to interpret or generate informal, context-dependent expressions. As a result, systems used for sentiment analysis, dialogue, and content moderation often struggle with Gen Z slang, where meaning depends heavily on tone, cultural context, and shared understanding (Sun et al., 2024). While prior work has explored slang detection and paraphrasing, relatively little research has focused on bidirectional translation between Gen Z slang and standard English, or on how such translation interacts with conversational properties like intent and emotion.

In this work, we address this gap by developing a framework that combines conversational understanding with stylistic text generation. We first explore dialogue act and emotion classification using transformer-based models trained on the DailyDialog dataset, allowing us to capture structured representations of intent (e.g., inform, question) and emotional tone. While these models perform reasonably well for dialogue acts, we observe clear limitations in emotion prediction due to class imbalance, with most outputs biased toward the “no emotion” label. Despite this, the classification component provides useful insight into how meaning is conveyed in conversational language.

Building on this, we develop a sequence-to-sequence translation system to convert between standard English and Gen Z-style language. Our approach uses a fine-tuned BART model trained on a combined multi-source corpus of slang expressions and sentence-level rewrites. To improve output quality, we generate multiple candidate translations and apply a reranking step that selects outputs which better preserve the original

meaning, intent, and tone. This allows us to move beyond surface-level fluency and consider whether generated text maintains the communicative function of the input.

Our contributions are threefold. First, we implement and evaluate dialogue act and emotion classification models, highlighting the impact of class imbalance on emotion prediction. Second, we develop a bidirectional Gen Z $\leftrightarrow$ standard English translation system, showing that BART provides strong performance for stylistic rewriting tasks. Third, we introduce a reranking-based analysis framework that evaluates translation quality through the lens of conversational consistency. Together, these components provide a step toward more robust NLP systems capable of handling informal, rapidly evolving language.

2. Related Work

Generational slang and NLP. Research on generational slang has primarily focused on detection and interpretation. Moore (2004) analyzed slang as a marker of generational identity. Sun et al. (2024) constructed the OpenSub-Slang benchmark and evaluated LLMs on slang detection, paraphrasing, and regional identification, finding that even GPT-4 struggles with inference and paraphrasing without task-specific fine-tuning. Bansal and Mamidi (2025) examined Gen Alpha slang evolution and evaluated AI translation systems (Google Translate, ChatGPT-4, Gemini, DeepSeek) on over 100 slang terms, finding significant failures rooted in cultural decontextualization and literal translation biases.

Dialogue act and emotion classification. The DailyDialog dataset (Li et al., 2017) provides multi-turn dialogues with per-utterance dialogue act and emotion annotations. Its emotion labels are heavily skewed toward "no emotion," a well-known challenge for training balanced classifiers on this dataset. We compare BERT (Devlin et al., 2019) with BERTweet (Nguyen et al., 2020), which is pre-trained on 850M English tweets and may better represent the informal register typical of Gen Z communication.

Style transfer and paraphrase generation. Style transfer in NLP involves rewriting text to change stylistic properties while preserving meaning. Related work includes formality transfer (Rao & Tetreault, 2018), Shakespeare-to-modern-English translation (Xu et al., 2012), and sentiment transfer. Our work adapts this paradigm to generational slang, using T5 (Raffel et al., 2020) and BART (Lewis et al., 2020) as translation backbones. BART's denoising pre-training objective, which involves reconstructing corrupted text, makes it particularly suited for paraphrase and rewriting tasks.

3. Data

3.1 DailyDialog

We use the DailyDialog dataset, which contains 13,118 multi-turn dialogues annotated with per-utterance dialogue act labels (inform, question, directive, commissive, dummy) and emotion labels (no emotion, anger, disgust, fear, happiness, sadness, surprise). To enable utterance-level modeling, we flatten each dialogue into individual rows, optionally including a short context window of the preceding 1–2 utterances.

After preprocessing, the dataset contains approximately 87,000 training utterances, with around 8,000 examples each for validation and test splits. A key characteristic of this dataset is the **strong class imbalance in emotion labels**: over 83% of utterances are labeled as "no emotion," while the remaining categories are sparsely represented. This imbalance plays a significant role in model performance, particularly for emotion classification.

3.2 Gen Z Slang Data

To train the translation model, we construct a corpus by combining multiple sources of Gen Z slang data.: (1) **MLBtrio/genz-slang-dataset** (HuggingFace), a lexicon of slang terms with descriptions and example

sentences; (2) **Programmer-RD-AI/genz-slang-pairs-1k**, containing 1,000 paired sentence rewrites between standard English and Gen Z style; (3) **thesherrycode/gen-z-slangs-translation**, a small set of paired short translations; (4) **GenZ_Translations.csv**, curated parallel sentence pairs; and (5) **GenZ_500_slang_terms.csv**, 500 slang terms with definitions, example sentences, and popularity indicators.

After normalization and deduplication, we create a set of bidirectional training pairs, allowing translation in both directions. Inputs are formatted with task-specific prefixes (**translate to Gen Z: / translate to standard English:**) to guide generation. This combined dataset captures both localized slang usage and broader stylistic patterns, which is important given that Gen Z communication often mixes standard language with informal expressions.

4. Methodology

4.1 Problem Formation

The goal of this work is to develop a system capable of translating standard English into Gen Z–style language while preserving both the original meaning and the conversational properties of the input. In addition to rewriting text, we aim to maintain properties such as conversational intent (e.g., whether a sentence is a question or statement) and emotional tone.

For example, a sentence such as *“I am sorry, I cannot make it tonight”* should be translated into a Gen Z equivalent such as *“my bad, I can’t pull up tonight.”* In this case, the output preserves both the intent (apology) and emotional tone (regret). This highlights two key challenges: (1) learning stylistic transformations and (2) ensuring pragmatic consistency between input and output.

To address this, we design a two-component pipeline that combines models for conversational understanding with a sequence-to-sequence model for style translation. The classification component provides insight into intent and tone, while the translation component generates stylistic rewrites.

4.2 Dialogue Act and Emotion Classification

To capture conversational intent and tone, we train models for dialogue act and emotion classification using the DailyDialog dataset, which contains multi-turn conversations annotated at the utterance level. Dialogue act labels include categories such as inform, question, directive, and commissive, while emotion labels are heavily skewed toward no emotion.

We preprocess the dataset by flattening dialogues into individual utterances, optionally including short context windows from preceding turns. This allows the models to operate at the utterance level while still incorporating limited conversational context.

We train both baseline and transformer-based classifiers. As a baseline, we use a TF-IDF representation with a Logistic Regression classifier. We then fine-tune pretrained transformer models, including BERT (bert-base-uncased) and BERTweet (vinai/bertweet-base), each with a linear classification head. BERTweet is included due to its pretraining on large-scale social media data, which more closely reflects informal language.

All models are trained using cross-entropy loss and evaluated using accuracy and macro F1-score to account for class imbalance. In practice, emotion prediction is strongly affected by this imbalance, with predictions frequently biased toward the no emotion label. While this limits performance on minority classes, the classifiers still provide useful signals for analyzing whether generated text preserves conversational intent and tone.

4.3 Translation Models

For the translation task, we use sequence-to-sequence transformer models trained on paired standard English and Gen Z–style text. The dataset is constructed by combining multiple sources of slang expressions and sentence-level rewrites, followed by normalization and deduplication. We structure the dataset in a bidirectional format, allowing the model to translate both from standard English to Gen Z and from Gen Z back to standard English. We experiment with two models:

- T5 baseline: We fine-tune t5-small using a text-to-text formulation, where all tasks are framed as sequence-to-sequence generation.
- BART model: We fine-tune facebook/bart-base, which is pretrained using a denoising objective that reconstructs corrupted text. This pretraining provides a strong inductive bias for tasks involving rewriting and paraphrasing.

Inputs are formatted with task-specific prefixes (e.g., “translate to Gen Z:” and “translate to standard English:”) to guide generation. Models are trained using teacher forcing with a cross-entropy loss objective. At inference time, the models generate stylistic rewrites rather than direct word-for-word translations, allowing for more flexible and natural outputs. In our experiments, BART consistently produces more fluent and stylistically appropriate outputs than T5, and is therefore used as the primary model in our final pipeline.

4.4 Candidate Generation and Reranking

To improve translation quality, we generate multiple candidate outputs for each input using beam search or similar decoding strategies. These candidates often vary in phrasing, tone, and stylistic features. Rather than relying on a single output, we compare multiple candidates and select those that better preserve the original meaning and conversational properties. We analyze each candidate using the dialogue act and emotion classifiers, comparing predicted labels between the input and generated output to assess consistency.

While we do not use a fully formalized scoring function, this process serves as a form of behavioral consistency checking. In practice, we prioritize candidates that maintain both semantic meaning and conversational intent, while also incorporating appropriate stylistic elements.

We use BERTweet as the primary model for analyzing reranking candidates. Although it does not significantly outperform BERT on the DailyDialog dataset, the reranking stage operates on Gen Z–style outputs, which are more similar to social media text. As a result, BERTweet’s pretraining is better aligned with this distribution.

This approach is motivated by the observation that a good translation should preserve not only semantic content but also communicative function. For example, a Gen Z expression conveying excitement should translate to a standard English sentence that reflects the same emotional tone, rather than a neutral or mismatched interpretation.

5. Results

5.1 Action/Emotion Classification

Model	Task	Accuracy	Macro-F1
BERT	Act	0.878	0.650
BERTweet	Act	0.887	0.656
BERT	Emotion	0.930	0.430
BERTweet	Emotion	0.930	0.328

Act classification shows reasonable performance for both models. The test set contains three of five act classes (inform, question, directive), with per-class F1 scores of 0.87, 0.92, and 0.80 respectively. BERTweet achieves a marginal improvement (+0.9% accuracy, +0.6% macro-F1).

Emotion classification reveals a fundamental challenge. Both models achieve 93% accuracy—a number that is almost entirely driven by correctly predicting the dominant "no emotion" class. The macro-F1 scores (0.430 for BERT, 0.328 for BERTweet) expose the real story: both models fail on minority emotion classes.

BERTweet actually performs *worse* than BERT on emotion macro-F1, a negative result we report transparently.

The "No Emotion" Problem. In qualitative testing with our Gradio demo, we observed a striking failure mode: even inputs with strong, obvious emotional content ("That makes me so angry!", "I'm so excited, this is amazing!") were consistently classified as "no emotion" by both models. This was not limited to Gen Z inputs—even standard English phrases with clear emotional markers received the "no emotion" label.

We identify four contributing factors: (1) **extreme class imbalance** (~83%+ "no emotion" in training) causes the model to learn that predicting the majority class is almost always correct; (2) **conservative annotation conventions** in DailyDialog, where only utterances with very explicit emotional markers receive emotion labels; (3) **standard cross-entropy loss**, which does not penalize the model for ignoring rare classes—the optimizer converges to a solution that minimizes overall loss by defaulting to majority-class predictions; and (4) for BERTweet specifically, **domain mismatch** between its tweet pre-training corpus and DailyDialog's formal scripted dialogue.

This has practical consequences for the reranking pipeline: when the emotion classifier assigns "no emotion" to nearly all inputs, the emotion component of the consistency score becomes uninformative, and reranking is driven almost entirely by the act classifier.

5.2 Translation Quality

Direction	Model	ROUGE-1	ROUGE-2	ROUGE-L	BERTScore-F1
Gen Z → Standard	T5	0.503	0.344	0.499	0.703
Gen Z → Standard	BART	0.806	0.685	0.803	0.976
Standard → Gen Z	T5	0.413	0.221	0.409	0.609
Standard → Gen Z	BART	0.680	0.530	0.667	0.948

BART achieves strong performance in both directions, with high BERTScore F1 values (0.976, 0.948) indicating robust semantic preservation. The directional asymmetry is expected: Gen Z → standard English is a more constrained task (typically one "correct" standard equivalent), while standard → Gen Z admits more creative variation and multiple valid outputs. This asymmetry is consistent with findings in other style transfer domains.

6. Discussion

6.1 Summary of Findings

Translation is the strongest component. BART achieves strong automatic metric scores for bidirectional Gen Z translation, with high semantic fidelity in both directions. The progression from T5 to BART demonstrates that model choice matters for this task, with BART's denoising pre-training providing clear advantages.

BERTweet does not substantively improve classification on DailyDialog. This lack of improvement reflects domain mismatch (tweets vs. formal scripted dialogue), not inherent model weakness. DailyDialog's language

register does not benefit from tweet-derived representations. However, we retain BERTweet in the reranking pipeline because the *inputs to reranking* are Gen Z slang—text that is stylistically much closer to tweets.

Emotion classification is the critical bottleneck. The "no emotion" dominance makes the emotion reranking signal uninformative for most inputs. The pipeline architecture is conceptually sound, but its full potential is gated on resolving the emotion classification limitation.

6.2 Addressing the Emotion Bottleneck

We propose four concrete directions for future work:

1. **Focal loss** (Lin et al., 2017): Down-weight the contribution of easy, high-confidence predictions (i.e., "no emotion") and focus training on hard, uncertain examples—precisely the emotional minority classes.
2. **Data augmentation for minority classes:** Use back-translation, LLM-based paraphrasing, or synonym substitution to synthetically expand the representation of emotional utterances in training.
3. **Multi-task training with external sentiment data:** Co-train the emotion classifier with sentiment analysis on Twitter sentiment corpora or other datasets with more balanced emotional distributions, providing additional supervisory signal.
4. **Gen Z emotion annotation:** For the specific use case of scoring Gen Z translations, the ideal solution is a dedicated emotion-annotated Gen Z text dataset. No such benchmark currently exists, representing an opportunity for future data collection.

These approaches are not mutually exclusive. A practical improvement path might combine class-weighted or focal loss with data augmentation as a near-term fix, while pursuing Gen Z emotion annotation as a longer-term investment.

7. Conclusion

In this work, we developed a pipeline for translating between standard English and Gen Z slang while preserving conversational intent and tone. Our approach combines dialogue act and emotion classification with a sequence-to-sequence translation model, using a fine-tuned BART model to generate candidate outputs and a reranking step to select translations that better align with the original input. Among all components, the translation model proved to be the strongest, with BART producing fluent and stylistically appropriate outputs in both directions. The progression from a T5 baseline to BART highlights the importance of model choice for this task. In contrast, improvements in classification were more limited, with BERTweet only slightly outperforming BERT, likely due to a mismatch between the DailyDialog dataset and the informal language used in Gen Z communication.

At the same time, our results highlight several important limitations. Emotion classification remains a key bottleneck, as strong class imbalance in DailyDialog causes models to default to the "no emotion" label for most inputs, reducing its usefulness in the reranking process. More broadly, the Gen Z translation dataset remains relatively small and limited in coverage, and the rapidly evolving nature of slang makes generalization difficult. Evaluation is also challenging: metrics such as ROUGE and BERTScore capture surface-level similarity but do not fully reflect stylistic nuance or cultural meaning. Future work could address these issues by improving dataset diversity, applying techniques such as data augmentation or class reweighting for emotion classification, and incorporating more robust human-centered evaluation methods. Overall, this work demonstrates both the potential and the current limitations of combining generation with conversational signals to model informal and rapidly evolving language.

Contributions

Ethan: I worked on the dialogue act and emotion classification part of the project, including training and evaluating the three models on the DailyDialog dataset. I also helped set up the evaluation pipeline for the translation system, including computing metrics like ROUGE and organizing the results for analysis. In addition, I contributed to refining the reranking approach and to all parts of the paper.

Jeevan: I was involved with the conceptualization of the project. I worked on formulating and creating the baseline act and emotion and doing some light fine-tuning on the model to get a good starting point. I also did the implementation of the Gradio app to coalesce the act and emotion model with the language translation, and created the UI for the website. I also contributed to writing and editing all sections of the paper.

Marvin: Involved in conceptualization of the project. Performed some light fine-tuning of the baseline act and emotion classification model. Experimented with act and emotion improvements over baseline; implemented BERTweet to draw parallels to more modern lexicon and grammar. Implemented and fine-tuned the baseline t5-small model for our initial bi-directional translation model. Added the act+emotion translation candidate rankings and consistency scores to the Gradio UI webpage. I also contributed to writing and editing all sections of the paper.

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Appendix: Interactive Demo

We built a Gradio web interface allowing users to enter text in either Gen Z slang or standard English, select a translation direction, view the best translation with behavioral preservation status, inspect the source utterance's act/emotion profile with full probability distributions, and browse all translation candidates ranked by consistency score. The demo launches with `share=True` for public access from Google Colab and provides a concrete, interactive demonstration of the full pipeline described in this paper.